

Mapping and Recognition of Body Movements on Another Person's Look-Alike Avatar

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Figure 1: The movement of three actors were captured and mapped onto the look-alike avatar of Actor A (leftmost actor).

Abstract

Research is needed to understand how users perceive the body movements of the look-alike avatar of a remote collaborator. To investigate, we captured three actors performing seven actions and mapped their body movements onto the look-alike avatar of one of the actors. Subsequently, we asked participants to identify the body movements of the actors rendered on the look-alike avatar. We found that participants were able to identify biological motions irrespective of the resemblance of the avatar to the controlling actor. The results of this study provide valuable insights into the accuracy of identifying body movements when mapped onto an avatar that bears no resemblance to the user. We discuss whether there are certain body movements that enables the user to accurately determine the person behind the avatar. The results have significant implications for trust and realism in collaborative virtual environments.

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CCS Concepts

• Human-centered computing → Virtual reality; • Computing methodologies → Perception.

Keywords

look-alike avatar, virtual characters, body movements, recognition

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1 Introduction

Avatars are representations of users in a virtual world, designed to enhance interaction in a virtual space. Recent advances in capture and render technology have enabled the rapid creation of avatars that resemble human subjects. Virtual interactions with realistic avatars with full face and body motion tracking have the potential to facilitate the generation of realistic interactions [Frampton-Clerk and Oyekoya 2022].

Motion Capture (MoCap) technology enables selectively decoupling and reconstruction of rendered behavior in collaborative virtual environments, a research phenomenon known as Transformed Social Interactions (TSI). TSI enables altering the nature of social interaction by giving interactants the tools to improve or worsen

interpersonal communication [Bailenson and Beall 2006]. Applying this phenomenon to look-alike avatars implies that users can potentially hijack the look-alike avatar of another person and pretend to be them. This leads to the question of whether interactants are able to see beyond the visual realism of the avatar. In other words, we explore whether users can identify the remote interactant or collaborators behind the look-alike avatar based on body movements. As such, this study explores the implications of embodying and controlling an individual's look-alike avatar by that same individual (faithful avatar representation) as against other individuals (unfaithful avatar representation).

2 Related Work

Johansson [Johansson 1973] concluded that visual perception of biological motion hinges on the interplay between information gained from movement patterns and the corresponding body shapes. To isolate motion information from the influence of form, a novel method utilizing a minimal set of bright markers placed on key joints to represent the motion of the living body was used. The results strongly suggest that the perception of motion like walking does not rely heavily on prior learning of specific motion patterns. Instead, findings point towards a more automatic and mechanical process of visual data processing underlying this recognition task. Another study [Cutting and Kozlowski 1977] also showed that the same set of light points is sufficient for the recognition of a specific individual in motion (walking). Narang et al. [Narang et al. 2017] conducted a study to identify factors that influence the perception of biological motion that are valuable to creating virtual avatars. Subjects were captured in an open space, with motions rendered on photorealistic avatars. Participants were asked to identify motions of themselves and other familiar individuals. Results showed that self-recognition was significantly higher for the photorealistic virtual avatars than with point-light representations, used in some studies. Users were more confident in their responses when identifying the motion presented on their avatar. McDonnell et al. [McDonnell et al. 2009] discuss an attribute of the body, specifically body type, and its influence on emotional perception. Results demonstrate relatively accurate recognition given that the best-performing actor's recognition rate was from 87% to 100%. The study also concluded that body type had no substantial effect on perception of emotions. The results of this study shows that captured body motion is as effective at depicting emotional body language and people's perception of emotional motion is very robust and unaffected by a character's physical appearance.

3 Methods

3.1 Participants and Materials

A total of sixty (12 females, 46 males, 1 non-binary, and 1 undisclosed gender) participants completed the anonymous online survey. The age ranges were: 18-24 (N=44); 25-34 (N=14); 35-44 (N=2) and 45-54 (N=0). Participants were recruited through emails to students, staff, and faculty at various institutions.

A look-alike avatar of one of the actors was generated using the Reallusion Character Creator v3.44 software, and the Headshot plugin v1.11. For this study, we used Optitrack (<https://optitrack.com>) MoCap System to animate the look-alike avatar in order to



Figure 2: Picture and Look-alike Avatar of Actor A

precisely match the timing and intensity of the real motion of the actors. The setup consisted of 10 Prime 13 cameras and five tripods. The capture area was set up in a room with a floor area of 4.5m long $\times$ 2m wide. A black mat was placed in the capture area to prevent reflections that can affect the tracking effectiveness of the cameras. The system came equipped with motion capture suits and retroreflective markers which were attached to different parts of the suit to enable tracking. Inside the capture area, two cameras each were mounted on a tripod. The cameras were connected to a switch, which uploaded all signals to a computer running Motive, the optical motion capture software. Since this study was focused on the whole body with passive fingers, forty-three (43) retroreflective markers were placed on the suits.

Three male actors volunteered for the motion capture sessions. All actors were non-professionals and accustomed to the motion capture setup and environment. In order to obtain highly comparable and similar motions for every actor, the actor whose look-alike avatar was generated (Actor A), acted out all the selected motions at the beginning, serving as the baseline for the other actors to mimic. Video recordings were taken on a mobile device during the motion capture session, to ensure that identical video and virtual stimuli could be achieved. Seven body movements (wave, walk, pray, head-thrust, fall, dance and angry) were selected amongst various motions acted out by the actors. Motion-captured animations from each actor was exported from Motive into Unity3D, where each set of motions was mapped onto the look-alike avatar. Video recordings of the avatar animations in Unity were also generated using the Unity recorder package.

3.2 Procedures and Measures

Using the video recordings of the avatar animations from Unity as well as the videos of the actors taken during the motion capture, an anonymous online survey was generated and distributed via Google Forms, which allowed for a diverse geographically distributed sample. Pre-experiment questions required users to provide their age and gender identification. Additionally, the images of Actor A and the corresponding look-alike avatar were presented at the beginning of the survey (as depicted in Figure 2) so that participants can rank the resemblance of the avatar on a scale of 1 (Not at all) to 10 (Very confident).

Subsequently, participants watched a 3x2 video with the actors presented on the top row and avatar animations presented on the bottom row. Participants had the option to replay the videos as many times as they deemed appropriate. The form had seven sections for each body movement, with each section containing video

representations of the actors and their corresponding avatar animation. Each section consisted of a 3x3 multiple-choice grid with each avatar animation presented in the rows and the actors presented in the columns. Participants were limited to one response per column, so one actor could not be matched with more than one avatar animation. For example, if a participant matched Actor A with Animation 3, then only Animations 1 and 2 can be selected for Actors B and C. The row order was shuffled randomly to reduce any confounding influence of the order and sequence effects such as learning or fatigue. Participants were able to change their opinions as needed before submitting their final responses, which could not be changed after submission. In each section, participants were asked to provide optional feedback on the body movement cues that they used for identification. In addition, participants were asked to rank on a likert scale of 1 (Least Realistic) to 5 (Most Realistic): (i) the overall intensity of the body movements of the actors; and (ii) the overall realism of the body animations on the avatars.

Post-experiment questions asked participants to rank how they feel the look-alike avatar resembles Actor A on a scale of 1 (least resemblance) to 10 (complete resemblance) based on the videos they watched. This enabled us to compare with the pre-test resemblance ratings. Participants were also asked to rank their confidence level on how they were able to accurately match the actors to their corresponding body movements, on a scale of 1 (Not at all) to 10 (Very confident). This enabled us to compare confidence with actual accuracy scores. Participants were also asked about familiarity i.e. if participants knew any of the actors, to gauge whether familiarity affects the matching accuracy. A final question asked for general feedback from the study.

4 Results

The experiment used a repeated measures (within-subjects) design with two independent variables: (i) (i) actors (A, B, and C); and (ii) body movements (wave, walk, pray, head-thrust, fall, dance, angry). We treat actors and body movements as categorical independent variables (factors), and accuracy as a dichotomous dependent variable and conducted a binomial logistic regression of the **accuracy** of the actors and body movements using SPSS Statistics 25 software. The binomial logistic regression was performed on the accuracy ratings to ascertain the effects of the actors and body movements on the likelihood that participants will accurately match the actors to their corresponding body movements on the look-alike avatar. Binomial logistic regression enables us to estimate the probability of participants being able to accurately match the avatar's body movement to the corresponding actor. If the estimated probability is greater than or equal to 0.5 (better than even chance), it is likely that cases can be correctly classified (i.e., predicted) from the independent variables. The logistic regression model was found to be statistically significant, ($\chi^2(8) = 87.900, p < 0.001$) and the model correctly classified 76% of cases. In other words, participants were generally able to match the actors' body movements to their corresponding avatar representation.

We treat the walking movement and Actor A as the reference categories for the independent variables, body movement, and actor respectively. For the body movements, statistical tests based on Likelihood-Ratio test results show that all motions ($\chi^2(6) =$

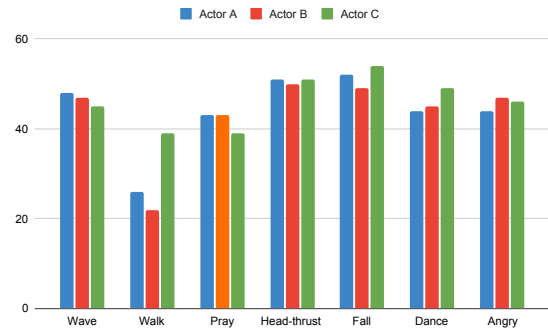


Figure 3: Accuracy frequency for each motion and actor

85.183, $p < 0.001$) added significantly to the model/prediction. In other words, participants were able to accurately match the actors' body movements to their corresponding avatar representation. The odds of accurately matching an actor's body movement to its corresponding avatar representation are greater for the angry (3.418 times), dance (3.525 times), fall (6.660 times), head-thrust (5.830 times), pray (2.436 times), and wave (3.756 times) in comparison to the walking body movements. For the actors, statistical test results show that neither Actor B ($p > 0.05$) nor Actor C ($p > 0.05$) added significantly to the model/prediction. The odds of accurately matching any of the actors' body movements to its corresponding avatar representations are similar based on statistical tests i.e. there was no statistically significant difference in odds between all actors. In other words, body movements were similarly identifiable regardless of the actor whose body movements were mapped to the look-alike avatar. Figure 3 shows the frequency of body movements matched accurately to the actors.

A non-parametric Friedman test of repeated measures was conducted to test the responses for the **perceived intensity** of the body movements of the actors. There were no statistically significant differences in participants' perceived intensity of the body movements of the actors ($\chi^2(6) = 11.928, p = 0.064$) with a median rank of 3 (walk, pray) and 4 (wave, head-thrust, fall, dance, angry). In other words, participants ranked the perceived intensity of the actors' body movements similarly. A non-parametric Friedman test of repeated measures was also conducted to test the responses for the **perceived realism** of the body animations on the avatars. There was a statistically significant difference in participants' perceived realism of the avatar animations ($\chi^2(6) = 42.637, p < 0.001$) with a median rank of 3 (walk, pray, dance) and 4 (wave, head-thrust, fall, angry). Dunn-Bonferroni post hoc tests found statistically significant differences ($p < 0.01$) between the praying movement and four of the avatar representations (dance, angry, head-thrust, fall). In other words, participants' perceived realism of the praying animations was generally lower.

We conducted a statistical test to compare the pre- and post-test **resemblance** ratings. The residual errors did not satisfy the tests of normality with a Shapiro-Wilks test, and the hypothesis of normality was rejected ($p < 0.05$). As such, a non-parametric Friedman test of repeated measures was conducted to test between the resemblance ratings. There was a statistically significant difference in participants' ratings of the avatar resemblance ($\chi^2(1) = 8.805, p < 0.003$) with a median rank of 6 (pre-test) and 7

(post-test). In order to compare participants' **perceived confidence** ratings with **actual accuracy**, the accuracy values were normalized from a maximum score of 21 to 10. The residual errors did not satisfy the tests of normality with a Shapiro-Wilks test, and the hypothesis of normality was rejected ($p < 0.05$). As such, a non-parametric Friedman test of repeated measures was conducted to test between participants' confidence ratings and actual accuracy. There were no statistically significant differences in participants' perceived confidence and their actual accuracy ($\chi^2(1) = 0.087, p = 0.768$) with a median rank of 8 (confidence) and 8 (accuracy).

5 Discussions

In this experiment, users judged actors' performance of actions to discern the realism of body movements, and whether users' perception of body tracking realism affects their assessment of overall realism. Participant recognition accuracy ratings for motion and corresponding actors were generally above average for all motions, indicating that a good number of participants correctly matched animations with the corresponding actors. This result suggests that users will be able to recognize when the person behind the avatar is not the person that looks like the avatar. The walk motion recorded the least recognition accuracy, in line with prior findings [Loula et al. 2005] that identity perception was found to be action dependent, as actors were better identified when they performed expressive actions such as dancing and boxing than typically constrained actions such as walking or running.

Participants were asked to rate the look-alike avatar's resemblance to Actor A (pre- and post-test), and the results demonstrated a trend exhibiting the uncanny valley effect. Statistical analysis shows that post-test resemblance ratings were statistically significantly higher than pre-test resemblance ratings. Participants rated the resemblance higher after watching the videos, showing that body movements add to the realism of look-alike avatars, in line with prior findings [Frampton-Clerk and Oyekoya 2022].

The change in resemblance of the look-alike avatars was a statistically significant improvement, and this was due to the added body movements. Approximately, two-thirds of the participants had not deviated past a ± 1 change, while most of the remaining participants had an improved perception (>2 change in score) of realism post-experiment. These results reinforce existing studies [Mousas et al. 2018] that articulate the significance of body movements on the perception of realism and demonstrate the effectiveness of motion-captured animations in virtual environments. Furthermore, the confidence of the participants were ranked similarly to the actual accuracy.

We used Actor A as the reference category because it served as a baseline for measuring the matching accuracy for the other actors, especially as the look-alike avatar was modeled based on the features of Actor A. The unfaithful representations (Actors B and C) had similar levels of matching accuracy as the faithful representation (Actor A).

Participants also rated the realism and intensity of the animations on a scale of 1 (least realistic) to 5 (most realistic). The results revealed that a majority of participants ranked all the animations with a score of 4-5 for both realism and intensity. Regarding the intensity ranking, the number of participants for this ranking demonstrated

little to no variation across all animations. More specifically, this scale range of 4-5 consisted of 45.9% - 68.8% of the participants. Additionally, 27.9% of participants changed the realism of the look-alike avatar to a score of 7-8, highlighting the significant impact of the animations.

Initial analysis of the responses and feedback regarding body movement cues that were employed for identification were aggregated and categorized based on body parts. Results indicate that the hands was the most dominant body part used for identification but we treat this potentially significant finding with caution and intend to explore it as future work.

Several limitations affect the generalization of the results of this study. The look-alike avatar was of a specific gender and race therefore, it is unclear how a diverse gender and race representation of actors may influence participants' choices. Furthermore, the participant's distribution is heavily skewed towards young male-identifying people. Future work will aim to run this study on an online platform such as MTurk for more diverse viewpoints. Finally, all actors used in this study were amateurs so it is not known whether these results will apply to professional actors.

6 Conclusions

In this study, we explored the ability of participants to accurately match seven body movements of three actors to their corresponding avatar animations. The look-alike avatar of Actor A was used as the baseline avatar onto which all actors' body movements were mapped. In effect, participants were judging whether they could identify a person based on their body movement, irrespective of the avatar's resemblance to the controlling actor. Results suggest that people are able to recognize the person controlling an avatar based on their body movements. Results also highlight the importance of body movements on the perceived realism of look-alike virtual characters. These results have positive implications for the use of look-alike avatars in meetings or social interactions. We envision a future where the remote avatar is a look-alike of the remote collaborator in a telepresence and social virtual environment. Trust is crucial in this scenario.

References

- Jeremy N Bailenson and Andrew C Beall. 2006. Transformed social interaction: Exploring the digital plasticity of avatars. In *Avatars at work and play: Collaboration and interaction in shared virtual environments*. Springer, 1–16.
- James E Cutting and Lynn T Kozlowski. 1977. Recognizing friends by their walk: Gait perception without familiarity cues. *Bulletin of the psychonomic society* 9 (1977), 353–356.
- Aisha Frampton-Clerk and Oyewole Oyekoya. 2022. Investigating the Perceived Realism of the Other User's Look-Alike Avatars. In *Proceedings of the 28th ACM Symposium on Virtual Reality Software and Technology*. 1–5.
- Gunnar Johansson. 1973. Visual perception of biological motion and a model for its analysis. *Perception & psychophysics* 14 (1973), 201–211.
- Fani Loula, Sapna Prasad, Kent Harber, and Maggie Shiffrar. 2005. Recognizing people from their movement. *Journal of Experimental Psychology: Human Perception and Performance* 31, 1 (2005), 210.
- Rachel McDonnell, Sophie Jörg, Joanna McHugh, Fiona N. Newell, and Carol O'Sullivan. 2009. Investigating the role of body shape on the perception of emotion. *ACM Trans. Appl. Percept.* 6, 3, Article 14 (sep 2009), 11 pages. <https://doi.org/10.1145/1577755.1577757>
- Christos Mousas, Dimitris Anastasiou, and Ourania Spantidi. 2018. The effects of appearance and motion of virtual characters on emotional reactivity. *Computers in Human Behavior* 86 (2018), 99–108. <https://doi.org/10.1016/j.chb.2018.04.036>
- Sahil Narang, Andrew Best, Andrew Feng, Sin-hwa Kang, Dinesh Manocha, and Ari Shapiro. 2017. Motion recognition of self and others on realistic 3D avatars. *Computer Animation and Virtual Worlds* 28, 3-4 (2017), e1762.